# Question Bank

# Internet of Things- PEC-CS-S 601

### 2 Marks Questions

1. Define Internet of Things (IoT).
2. What is M2M communication?
3. List any two sources of IoT.
4. What is meant by IoT architecture?
5. Define sensor.
6. What is RFID technology?
7. What is wireless sensor network?
8. Define data aggregation.
9. What is Arduino IDE?
10. What is virtual memory?
11. What is IoT conceptual framework?
12. Define actuator.
13. What is participatory sensing?
14. Name any two IoT hardware platforms.
15. What is node discovery?
16. Define routing protocol.
17. What is MAC protocol?
18. What is Arduino board?
19. What is data dissemination?
20. What is ease of designing in IoT?
21. Give one example of IoT architecture implementation.
22. Compare ARM Cortex architecture with Intel Galileo for IoT hardware, focusing on power efficiency and sensor integration.
23. What do you mean by MAC protocol?
24. Briefly classify the sensors.
25. What are different challenges of IoT?

26. Define Embedded computing.
27. Mention any two differences between IoT and M2M communication.
28. How does Arduino IDE handle advanced programming for hybrid IoT platforms like Raspberry Pi, including code for actuator control and wireless medium access?
29. What communication technology can be used between streetlights for coordination?
30. What are the advantages of sensor network?

**5 Marks Questions**

1. Explain the vision and concept of IoT.
2. Describe design principles for connected devices.
3. Explain different types of sensors used in IoT.
4. Discuss embedded platforms for IoT with examples.
5. Explain wireless medium access issues in IoT networks.
6. Discuss various IoT applications such as smart metering and home automation.
7. Explain sources of IoT with suitable examples.
8. Discuss communication technologies used in IoT.
9. Explain RFID and its applications.
10. Describe node deployment and discovery in IoT.
11. Explain data enrichment and consolidation in IoT systems.
12. Discuss challenges in IoT design.
13. Describe the architectural design considerations for connected IoT devices.
14. Discuss the various communication technologies used in IoT systems with suitable examples.
15. What is Arduino IDE? Explain programming of the Arduino for IoT.
16. Narrate the concept of sensor deployment and node discovery.
17. Examine wireless medium access issues in IoT, including a survey of MAC protocols, routing protocols, and the role of low-duty cycle concepts in sensor networks.
18. Design an advanced IoT system for smart city automation, explaining key features of mobiles/tablets, data communication with hardware, and major security challenges.

## 10 Marks Questions

1. Explain IoT architecture and technologies behind IoT in detail.
2. Discuss design challenges in IoT including security and development issues.
3. Explain IoT system layers and communication technologies in detail.
4. Describe IoT applications in smart cities and E-health with examples.
5. Explain IoT conceptual framework and architectural view in detail.
6. Discuss IoT applications in automotive systems and smart cities.
7. Explain embedded computing basics and IoT hardware platforms.
8. Describe security challenges in IoT with suitable examples.
9. Differentiate digital sensors, actuators, and embedded hardware for IoT, including neat sketches and analysis of RFID versus wireless sensor networks.
10. Critically analyze IoT/M2M system layers, design standardization, and their implications for architectural views, including ease of design, affordability, and open innovation.
11. In IoT-based remote patient monitoring, wearable devices send data to hospitals for real-time health analysis. Explain how the IoT framework and architecture operate in this scenario. Discuss the enabling technologies, nature of IoT data sources, and the use of M2M communication in emergency alert generation.
12. Evaluate the development, security, and design challenges in IoT, focusing on data enrichment, consolidation, and predictive/prescriptive analytics for applications like home automation.